# Calories Burnt Prediction

**Linear Regression with Gradient Descent and OLS**

## 1. Introduction and Dataset

This project predicts calories burnt during exercise using multivariate linear regression. Gradient Descent (GD) is implemented from scratch and compared with the closed-form Ordinary Least Squares (OLS) method. A second comparison checks whether adding engineered features improves predictions.

Only the two provided dataset files were used. Each contains 15,000 records. The exercise file contains User_ID, Gender, Age, Height, Weight, Duration, Heart_Rate, and Body_Temp. The calories file contains User_ID and Calories. Merging on the unique User_ID produced 15,000 rows and nine columns, with all exercise records matched.

## 2. Data Preprocessing and Cleaning

All numerical columns had appropriate numeric types. There were no missing values, duplicate rows, or repeated User_ID values, so filling missing values and removing duplicates were unnecessary. Gender contained 7,553 female and 7,447 male records; these were encoded as 0 and 1 respectively. User_ID was removed before modeling.

An 80/20 split with random_state=42 produced 12,000 training records and 3,000 test records. The same split was used for every model. Predictors were standardized using training means and standard deviations only; these same values were applied to the test set. Calories remained in its original units. A column of ones was added for the intercept.

Outliers were checked using boxplots and the IQR rule: values below Q1 - 1.5 × IQR or above Q3 + 1.5 × IQR were flagged, where IQR = Q3 - Q1. This initial inspection did not change any observations.

| Column | IQR flags | Observed range |
|---|---|---|
| Age | 0 | 20-79 years |
| Height | 14 | 123-222 cm |
| Weight | 6 | 36-132 kg |
| Duration | 0 | 1-30 minutes |
| Heart_Rate | 1 | 67-128 |
| Body_Temp | 369 | 37.1-41.5°C |
| Calories | 4 | 1-314 |

Flagged records were retained because an IQR flag alone does not prove an error. Body_Temp flags were at the lower end of its distribution. Measurement details were unavailable, so the temperature range remains a data-quality limitation. Counts are per column and can overlap between rows.

## 2. Data Cleaning: Boxplot Evidence

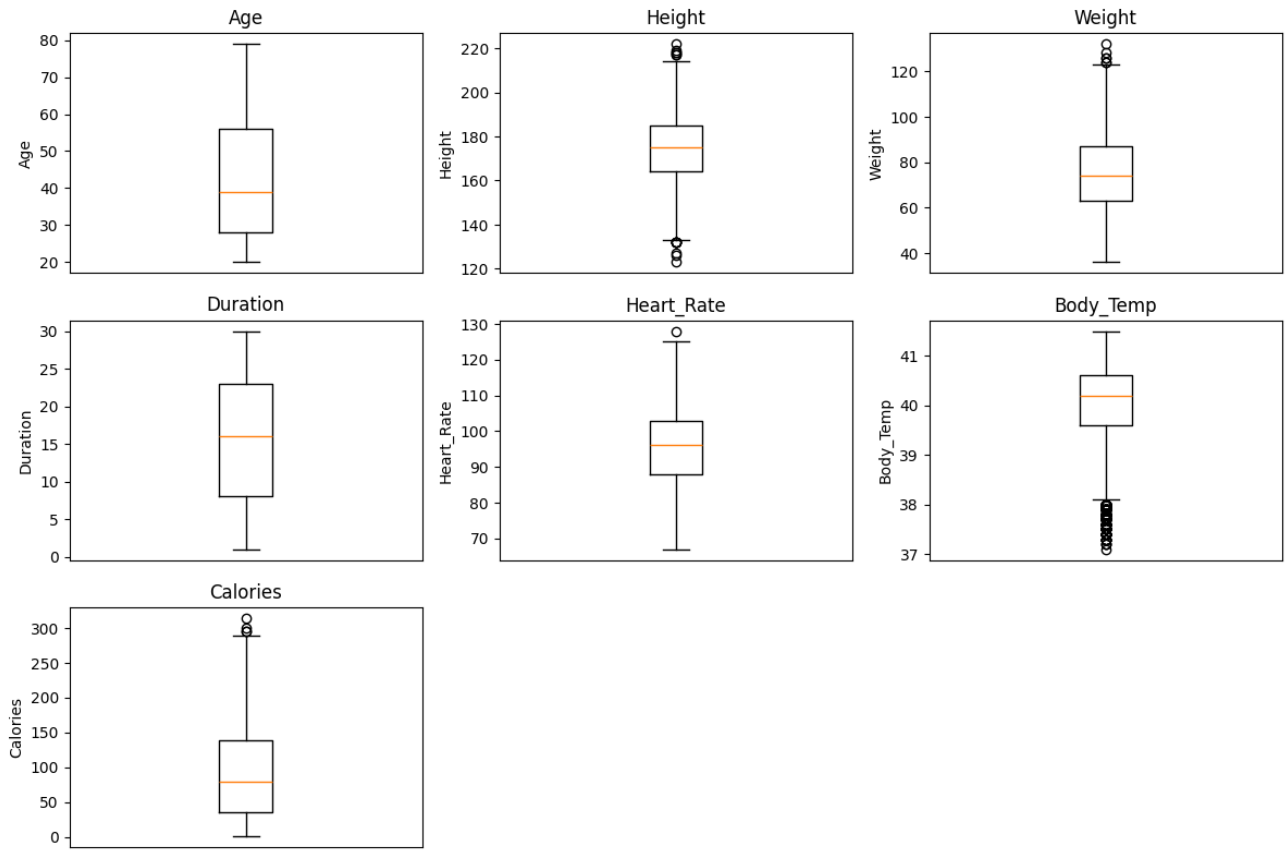



Figure 1. Boxplots of numerical variables; User_ID is excluded.

## 3. Linear Regression Assumption: Linearity

Linearity was examined before model training using scatterplots of the seven original predictors against Calories in the training set (Figure 2). Gender is binary, so it forms two groups rather than a continuous trend.

Duration and Heart_Rate show strong positive relationships with some curvature. Body_Temp shows a pronounced upward curve. Age, Height, and Weight show considerable scatter and weaker individual patterns.

Straight-line relationships are therefore not fully supported for all raw predictors. Raw linear terms can miss curved patterns and produce systematic prediction errors. Individual scatterplots are an initial check; they do not fully establish the combined multivariate relationship.

## 4. Feature Engineering

| Added feature | Formula and justification |
|---|---|
| BMI | Weight / (Height / 100)². Represents weight relative to height. |
| Duration_Heart_Rate | Duration × Heart_Rate. Allows the duration effect to vary with heart rate. |
| Duration_Squared | Duration². Allows curvature in the duration relationship. |

All seven original predictors were retained. Adding these three features produced ten predictors. The formulas were applied separately to each split without estimating anything from test data. Polynomial and interaction terms still produce a model that is linear in its coefficients.

# 3. Linearity: Training-Set Scatterplots

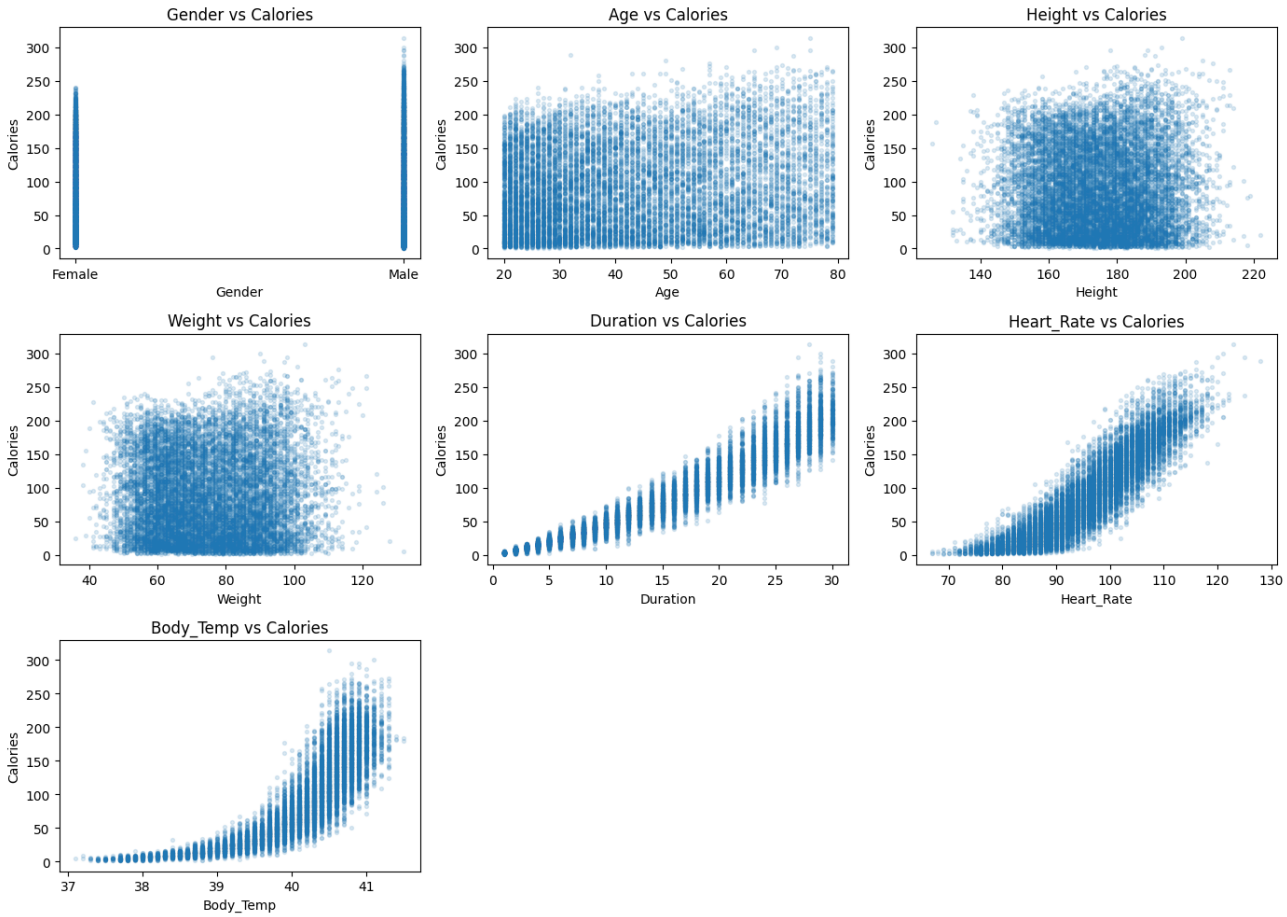



Figure 2. Original predictors against Calories, using training records only.

# 5. Model Training and Evaluation Methods

For m observations, X is the design matrix including the intercept column, y is actual Calories, and w is the coefficient vector. GD starts with all coefficients at zero.

**Prediction:** $y_hat = Xw$
**Cost:** $MSE = (1/m)\ \Sigma(y_hat - y)^2$
**Gradient:** $(2/m)\ X^T(Xw - y)$
**Update:** $w_new = w - \alpha \times gradient$

Each learning rate was allowed 10,000 updates. Runs stopped when the cost became non-finite or exceeded $10^{20}$. This safeguard detects numerical divergence, not convergence. Costs were recorded before each update for the learning curves; final costs were recalculated after training.

**Evaluation:** $MAE = mean(|y - y_hat|)$; $RMSE = sqrt(mean((y - y_hat)^2))$; $R^2 = 1 - \Sigma(y - y_hat)^2 / \Sigma(y - mean(y))^2$. Test metrics use the test-set target mean in $R^2$. Lower MAE/RMSE and higher $R^2$ indicate better predictions.

# 6. Gradient Descent Results

| Learning rate | Updates | Final training MSE | Time (s) | Outcome |
|---|---|---|---|---|
| 0.001 | 10,000 | 89.41637 | 2.496217 | Completed |
| 0.01 | 10,000 | 75.87047 | 1.807375 | Completed |
| 0.1 | 10,000 | 60.17738 | 1.762258 | Completed |
| 0.5 | 15 | $1.264285 \times 10^{20}$ | 0.002721 | Diverged |
| 1.0 | 10 | $5.638661 \times 10^{21}$ | 0.001775 | Diverged |

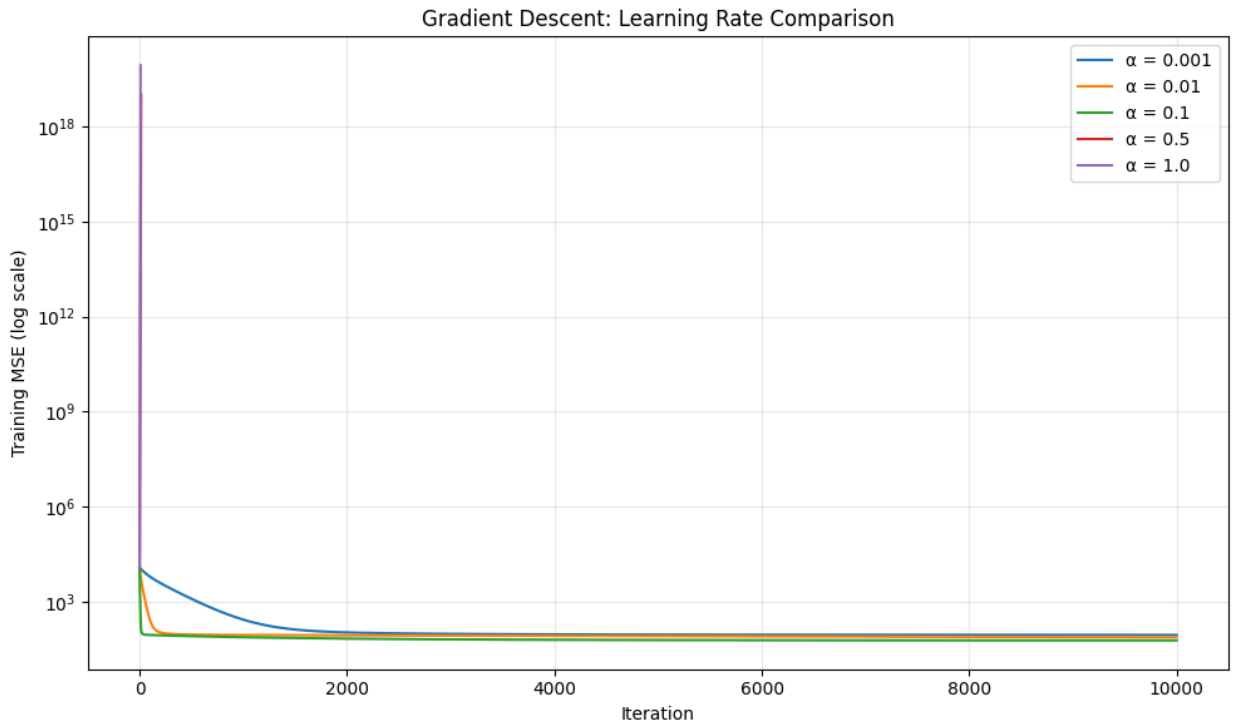



Figure 3. All five learning rates on one graph. The vertical axis is logarithmic.

The 0.001 run decreased slowly. The 0.01 run decreased faster initially, but 0.1 reached the lowest training MSE after 10,000 updates and was selected without using test results. Rates 0.5 and 1.0 diverged rapidly. No clear cost oscillation is visible for the stable runs at this plot scale. Completion of 10,000 updates does not prove full convergence.

| Learning rate | Test MAE | Test RMSE | Test $R^2$ |
|---|---|---|---|
| 0.001 | 6.872707 | 9.454369 | 0.977172 |
| 0.01 | 6.309996 | 8.731663 | 0.980528 |
| 0.1 (selected) | 5.836811 | 7.898100 | 0.984069 |

Diverged runs are not treated as usable predictive models; test metrics are omitted for those runs. Final coefficients for the selected GD model are reported alongside OLS on the next page.

## 7. OLS / Normal Equation

OLS minimizes SSE = $(y - X\beta)^T(y - X\beta)$. Differentiating gives $-2X^T(y - X\beta)$. Setting this to zero gives $X^TX\beta = X^Ty$, so $\beta = (X^TX)^{-1}X^Ty$ when $X^TX$ is invertible. NumPy solve was used to solve these normal equations without explicitly forming the inverse.

OLS used the same standardized features as GD. Its test MAE was **5.849070**, RMSE was **7.899607**, and $R^2$ was **0.984063**. Training took **0.001780 seconds** in the recorded Colab run.

## Estimated Coefficients and GD Comparison

| Term | GD: scaled inputs | OLS: scaled inputs | OLS: original units |
|---|---|---|---|
| Intercept | 89.647000 | 89.647000 | 323.949113 |
| Gender | -0.074293 | -0.077156 | -0.154314 |
| Age | 8.625105 | 8.624991 | 0.507878 |
| Height | -29.429273 | -29.541822 | -2.077946 |
| Weight | 37.754166 | 37.895718 | 2.529605 |
| Duration | -56.452535 | -60.806059 | -7.321125 |
| Heart_Rate | 0.777854 | -0.323546 | -0.033685 |
| Body_Temp | 0.070428 | 0.188816 | 0.243908 |
| BMI | -11.417478 | -11.458341 | -7.351157 |
| Duration_Heart_Rate | 114.305439 | 121.224487 | 0.130761 |
| Duration_Squared | 1.792764 | 0.052107 | 0.000196 |

Original-unit coefficients were recovered as $b_j = \beta_j / \text{training_std}_j$ and $b_0 = \beta_0 - \Sigma(\text{training_mean}_j \times b_j)$. Both coefficient representations give the same OLS predictions.

## Interpretation of Three OLS Coefficients

**Age:** One additional year is associated with about 0.508 more predicted calories, holding other predictors constant.
**Gender:** With female = 0 and male = 1, males have about 0.154 fewer predicted calories than females with the other predictors fixed.
**Body_Temp:** A 1°C increase is associated with about 0.244 more predicted calories, holding other predictors constant.

These are conditional model associations, not causal effects. Duration and Heart_Rate also enter engineered terms, so their individual coefficients do not represent their complete effects. Correlated predictors make individual coefficients harder to interpret.

## 8. GD vs. OLS

| Model | Training MSE | Test MAE | Test RMSE | Test $R^2$ | Time (s) |
|---|---|---|---|---|---|
| GD | 60.177383 | 5.836811 | 7.898100 | 0.984069 | 1.762258 |
| OLS | 60.079736 | 5.849070 | 7.899607 | 0.984063 | 0.001780 |

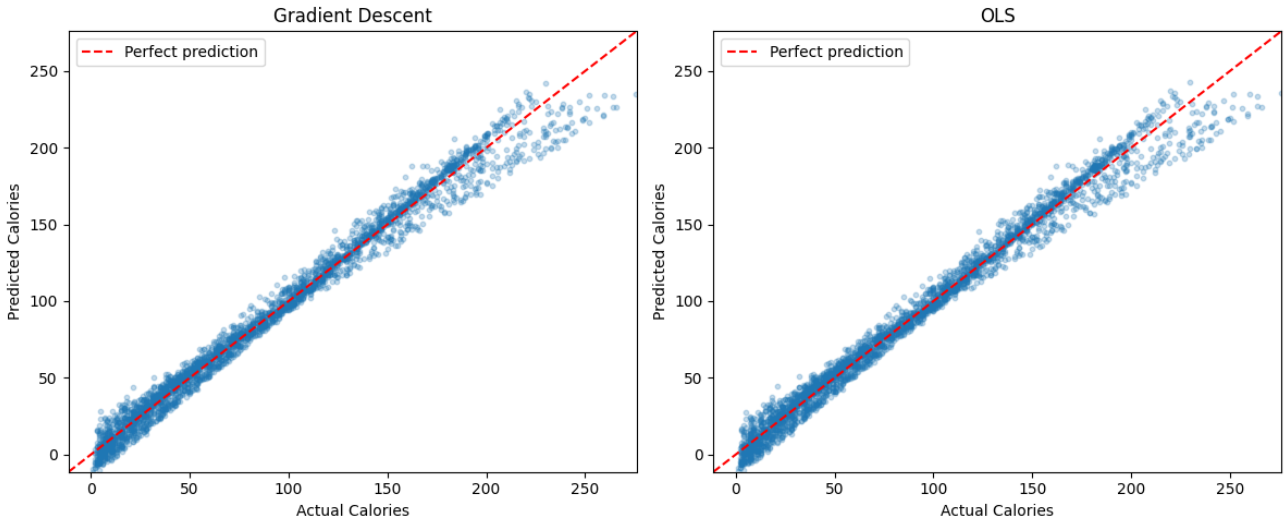



Figure 4. GD and OLS predictions on the same test set.

Both methods achieved nearly identical test accuracy. GD had slightly lower test errors, but OLS achieved lower training MSE. OLS minimizes training squared error; this does not guarantee the lowest test error.

The intercept and several coefficients were close, but differences remained in Duration, Heart_Rate, and the engineered duration terms. GD had not fully reached the OLS solution after 10,000 updates. Correlated predictors can allow different coefficients to yield similar predictions.

OLS was more practical for this dataset of 12,000 training records and ten predictors: it gave comparable accuracy with much shorter measured training time. Timings cover fitting, exclude shared preprocessing, and depend on the runtime environment. OLS was selected for the feature comparison on this basis. Both models underpredicted some high-calorie observations.

## 9. Raw vs. Engineered Features

Model A was trained with OLS using the seven original predictors. Model B reused the OLS model with all ten predictors. Both used identical training/test records and the same training-based scaling for shared features.

| OLS model | Predictors | Test MAE | Test RMSE | Test $R^2$ |
|---|---|---|---|---|
| A: Raw | 7 | 8.408541 | 11.313070 | 0.967313 |
| B: Raw + engineered | 10 | 5.849070 | 7.899607 | 0.984063 |

# 9. Feature Comparison: Prediction Plots

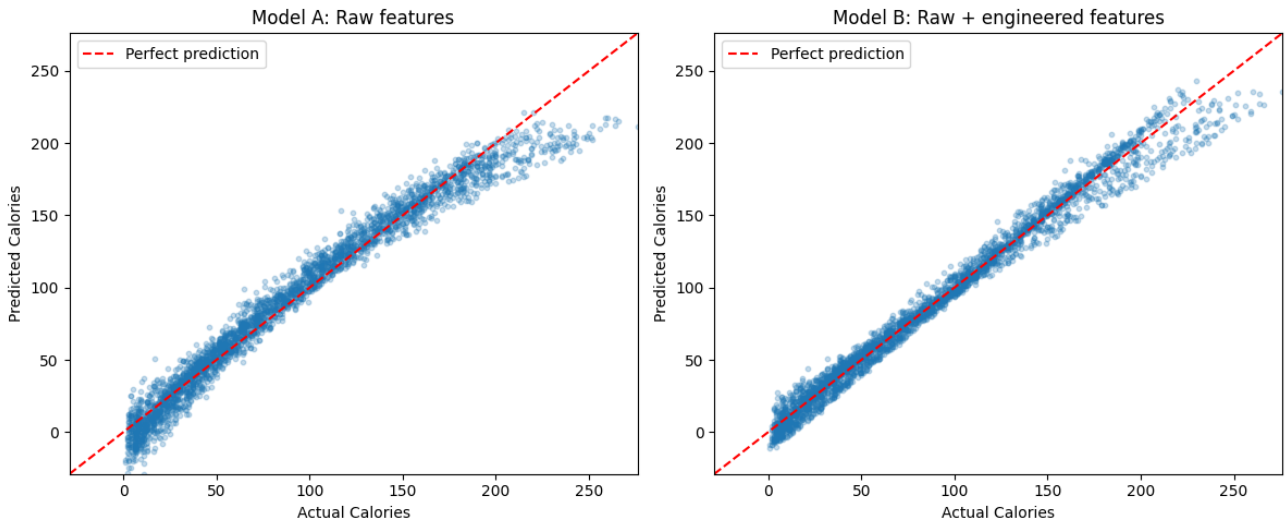



Figure 5. Adding engineered features brings predictions closer to the perfect-prediction line.

Feature engineering reduced MAE by approximately 30.4% and RMSE by 30.2%. $R^2$ increased from 0.9673 to 0.9841. The added features represent relative body size, an interaction between duration and heart rate, and duration curvature. Together, they improved performance on this split. This experiment does not measure each feature's individual contribution.

# 10. Conclusion and Limitations

Both GD and OLS predicted calories burnt successfully. Of the tested learning rates, 0.1 gave the lowest training MSE after 10,000 updates; 0.5 and 1.0 diverged. OLS reached a lower training MSE and trained much faster while giving almost identical test performance. Adding three engineered features substantially improved the raw-only OLS model.

Results are based on one train/test split, so performance on other samples may differ. Some high-calorie observations remain underpredicted, and a few predictions are negative even though negative calories are not meaningful. No prediction clipping was applied. Unusual measurements could not be verified without measurement details. Correlated predictors complicate coefficient interpretation, and GD had not fully reached the OLS solution within its update budget.